

Report on Big Data and Python Practice

Today's session introduced me to the basics of Python programming and Big Data. We worked with a dataset containing information about 20 students, such as their names, ages, gender, programs, districts, attendance, and scores. We used Python to create variables, work with strings and lists, access information from dictionaries, and calculate basic statistics from the student dataset.

From the analysis, I learned that the 20-student dataset is not Big Data because it is too small and can be easily stored and processed using a normal computer. We also discussed the five V's of Big Data: **Volume**, **Velocity**, **Variety**, **Veracity**, and **Value**. The dataset clearly does not have enough **Volume** because it contains only 20 students. It also has very little **Velocity** because the data is not being collected or updated continuously in real time.

One thing that surprised me during the session was how Python handles different types of data. For example, when we write `"3" + "4"`, Python

gives `"34"` instead of `7` because the numbers are stored as strings. This helped me understand the importance of knowing the data type when writing Python programs. Overall, the session helped me understand the connection between Python, data analysis, and Big Data.